

TECHNICAL DOCUMENTATION

MediCare Pharmacy Management System

A full-stack enterprise platform for digitalising and automating daily pharmacy operations, from single-branch stores to multi-branch chains.

Document type	System overview & reference guide
Architecture	Three-tier — Angular · Spring Boot · PostgreSQL
Access model	Role-based — Admin, Manager, Pharmacist, Cashier
Deployment	Dockerised, single-command startup
Version	1.0
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Table of Contents

01	Overview	3
02	Key Features	3
03	Technology Stack	6
04	Architecture	6
05	User Roles & Permissions	7
06	Quick Start — Local Development	8
07	Production Deployment	8
08	Summary	9

SECTION 01

Overview

The MediCare Pharmacy Management System is a full-stack enterprise web application designed to digitalise and automate daily operations of pharmacies, from single-branch stores to multi-branch chains.

It provides role-based access for administrators, managers, pharmacists, and cashiers, with modules covering inventory, point of sale (POS), prescriptions, patients, purchasing, alerts, reporting, and more.

The system offers a modern, responsive user interface, a secure REST API backend, and can be deployed anywhere — on-premises or in the cloud.

SECTION 02

Key Features

Authentication & Multi-Role Access

- Secure login with JSON Web Tokens (JWT)
- Four predefined roles: Admin, Manager, Pharmacist, Cashier
- Each role sees only the modules and data it is authorised for, enforced in both the frontend sidebar and the backend API
- Users can update their own profile and change their password

Multi-Branch Management

- Create and manage an unlimited number of pharmacy branches
- Every user is assigned to a specific branch
- All inventory, sales, and operations are automatically scoped to the user's branch
- Admins can view and manage all branches and users across the entire organisation

Inventory & Batch Tracking

- Medication catalogue with barcode, generic name, category, and unit
- Stock management via batches: each batch is linked to a medication, supplier, and branch, and stores quantity, expiry date, purchase price, and selling price
- FEFO (First-Expiry-First-Out) batch selection in POS to minimise waste
- Low-stock and expiry alerts generated automatically by a scheduled task
- Soft-delete of batches (quantity set to zero) preserves audit history while removing items from active stock views

Point of Sale (POS)

- Search medications by name or scan barcodes using a camera (ZXing) or a physical USB scanner
- Real-time batch selection with available stock, expiry date, and selling price
- Cart management: add, remove, and adjust quantities
- Optionally link a patient and/or a prescription to a sale
- Apply discounts and select payment methods — Cash, Credit Card, Insurance
- Sale completion automatically deducts stock from the selected batch
- PDF invoice generation for each sale

Prescriptions & Patients

- Maintain a complete patient database with contact details
- Create digital prescriptions with multiple medication items, dosage, duration, and doctor details
- Prescriptions can be attached to POS sales, ensuring the correct medications are dispensed

Dashboard & Analytics

- Real-time summary cards: total branches, users, sales, low-stock items, and active alerts
- Interactive charts built with ECharts, including a sales trend line chart (revenue and transaction counts over 7, 30, or 90 days) and a top-medications bar chart by revenue
- Recent sales table and active alerts list
- Skeleton loaders for a smooth loading experience

Reports

- Sales summary — total sales, total revenue, average sale value

- CSV export for medications, patients, and sales reports, allowing easy data analysis in Excel

Alerts & Notifications

- Automatic detection of low-stock batches (quantity ≤ 10) and approaching expiries (within 30 days)
- Alerts are displayed on the dashboard and a dedicated alerts page
- Email notifications are sent to all admins and managers when new alerts are generated, requiring SMTP configuration

Audit Logs

Every create, update, and delete operation on key entities — medications, suppliers, users, and others — is logged with the entity name and ID, the action type (CREATE / UPDATE / DELETE), a description of the change, and a timestamp with the user who performed it. Audit logs are viewable only by administrators.

Barcode Scanning

- Medications can be assigned a barcode value
- POS supports two scanning methods: a camera scanner using the ZXing library, and manual input for USB barcode scanners that emulate keyboard input
- Scanning a barcode instantly selects the corresponding medication

Modern User Interface

- Fully responsive, professional design with a fixed dark sidebar, smooth animations, and skeleton loaders
- Dark mode toggle, with the preference saved across sessions
- Toast notifications for success, error, and warning feedback
- Interactive charts with dark-mode-aware colour theming

Email Notifications

- Integrated with Spring Boot Mail
- Sends alerts for low-stock and expiry items to all admin and manager email addresses

Dockerised Deployment

- Entire stack is containerised with Docker and Docker Compose
- Three services: PostgreSQL, Spring Boot backend, and Angular frontend served by Nginx
- One-command startup via `docker compose up -d`

SECTION 03

Technology Stack

Layer	Technology
Backend	Java 17, Spring Boot 3.2, Spring Security, Spring Data JPA (Hibernate)
Frontend	Angular (standalone components), TypeScript, Bootstrap 5, Font Awesome
Database	PostgreSQL 16
Authentication	JWT (JSON Web Tokens) with BCrypt password hashing
PDF Generation	iText 8
Charts	ECharts, via ngx-echarts
Barcode Scanning	ZXing — @zxing/ngx-scanner and @zxing/library
Build Tools	Maven (backend), Angular CLI (frontend)
Containerisation	Docker, Docker Compose
Email	Spring Boot Mail (JavaMailSender)

SECTION 04

Architecture

The application follows a classic three-tier architecture:

ANGULAR FRONTEND → SPRING BOOT REST API → POSTGRESQL

- Frontend communicates with the backend exclusively via RESTful JSON APIs
- A JWT interceptor attaches the token to every request automatically
- The backend enforces role-based access with `@PreAuthorize` and a `SecurityFilterChain`
- Layered design: Controller → Service → Repository → Entity
- A global exception handler returns clean JSON errors
- Hibernate automatically creates and updates tables during development

SECTION 05

User Roles & Permissions

Module	Admin	Manager	Pharmacist	Cashier
Dashboard	Yes	Yes	Yes	Yes
Profile	Yes	Yes	Yes	Yes
Branches	Yes	—	—	—
Users	Yes	—	—	—
Suppliers	Yes	Yes	—	—
Medications	Yes	Yes	Yes	—
Inventory & Purchases	Yes	Yes	—	—
Point of Sale	Yes	Yes	Yes	Yes
Patients	Yes	Yes	Yes	—
Prescriptions	Yes	—	Yes	—
Reports	Yes	Yes	—	—
Alerts	Yes	Yes	Yes	—
Audit Logs	Yes	—	—	—

SECTION 06

Quick Start — Local Development

1 Start the database

```
docker compose up -d postgres
```

2 Start the backend

```
cd backend && ./mvnw spring-boot:run
```

3 Start the frontend

```
cd frontend && ng serve
```

4 Access the application

Open `http://localhost:4200`

Login — `admin@pharmacy.com` / `admin123`

SECTION 07

Production Deployment

The system is fully Dockerised. To run everything with a single command:

```
docker compose up -d --build
```

The frontend is served by Nginx on port 80, with API requests proxied to the backend on port 8080. Database data is persisted in a Docker volume.

SECTION 08

Summary

The MediCare Pharmacy Management System is a production-ready, feature-complete platform that modernises pharmacy operations. Its clean architecture, modern UI, robust security, and extensive functionality make it suitable for real-world pharmacy environments, whether a single store or a large chain.

The system is easy to deploy, easy to maintain, and designed with scalability in mind.